

Conformal Coverage Under Quantisation: A Case Study on Edge-Deployed YOLOv8n

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Abstract—Post-training quantisation is a standard technique for deploying object detectors on CPU-class edge hardware, yet its effect on conformal coverage in edge-deployed object detection remains underexplored. This paper presents, to our knowledge, the first empirical investigation of split conformal prediction (CP) applied to a YOLOv8n detector exported via Intel OpenVINO at FP32, FP16, and INT8 precision, evaluated on subsets of COCO128 and COCO val2017. We design a 24-condition experiment (3 precisions $\times$ 8 input conditions: clean, three levels of Gaussian blur, three JPEG compression qualities, and additive Gaussian noise) and show that, while the FP32 model achieves an empirical coverage rate of 0.9165 on clean test data, the FP16 and INT8 quantised models fall below the nominal 90% target (0.8929 and 0.8976 respectively). Under degradation, FP32 violates coverage in 1 of 8 conditions, compared to 6/8 for FP16 and 3/8 for INT8. We further show that the convergence of mean nonconformity scores across precisions under degradation mirrors the confidence convergence phenomenon reported in our earlier calibration study, and provide an empirical reliability interpretation of how quantisation and input degradation jointly erode empirical conformal coverage over matched detections. We note that coverage is measured over matched detections under a fixed IoU threshold and LAC scoring protocol, and does not constitute a blanket end-to-end detection reliability guarantee. Our results demonstrate that post-training quantisation should be treated as a model change from the standpoint of conformal prediction: coverage should be re-validated empirically after compression and validated under realistic deployment conditions.

This paper addresses four research questions:

- RQ1 Do conformal thresholds differ significantly across FP32, FP16, and INT8 precision levels?
- RQ2 Do FP16 and INT8 models maintain the nominal 90% conformal coverage target on clean test data?
- RQ3 How does input degradation affect conformal coverage across precision levels, and which precision is most robust?
- RQ4 Can the coverage violations observed be connected to the confidence-convergence phenomenon reported in [1]?

Index Terms—conformal prediction, quantisation, YOLOv8n, OpenVINO, edge computing, object detection, uncertainty quantification, coverage target, distribution shift

I. INTRODUCTION

Edge-deployed object detectors are now embedded in autonomous vehicles, urban surveillance infrastructure, and industrial monitoring pipelines, where models must run under strict latency and memory budgets on CPU-class hardware. In this setting, post-training quantisation has become a standard technique for compressing models by reducing numerical precision from 32-bit floating point (FP32) to half-precision (FP16) and 8-bit integer (INT8) formats, yielding substantial throughput gains without retraining [18]–[20]. While a large body of work has demonstrated that carefully calibrated INT8 pipelines can preserve mean Average Precision (mAP) within a few percentage points of FP32 baselines [13], [17], evaluations overwhelmingly focus on accuracy and frames-per-second metrics.

For reliability-sensitive deployments, these metrics are incomplete. Downstream decision-making components rely on detector confidence scores to apply thresholds and business logic; a model that retains mAP but becomes systematically overconfident after quantisation can be more dangerous than a slightly less accurate but honestly calibrated one, because high-confidence false positives and false negatives are more likely to pass unnoticed through monitoring layers [15]. Our previous work characterised this risk through confidence calibration analysis (Expected Calibration Error, reliability diagrams) and discovered a *confidence convergence phenomenon*: a 0.382 confidence gap between FP32 and INT8 models on clean data collapses to under 0.034 under any degraded input condition (hereafter *Phase 1*, [1]).

Conformal prediction (CP) offers a principled, complementary perspective. CP constructs prediction sets with a user-specified marginal coverage target $1 - \alpha$ under minimal assumptions, typically exchangeability of calibration and test data [2]. Split conformal prediction (SCP) is particularly attractive in practice as it can be applied post-hoc to any black-box model using a held-out calibration set [3], [4]. Recent work has begun applying CP to object detection via two-step conformal schemes that handle class labels and bounding-box uncertainty simultaneously [5] [6] [7]. Meanwhile, several studies have demonstrated that conformal guarantees can de-

grade under distribution shift, including image corruption and domain shift [11], [9]. The interaction between *quantisation* and conformal coverage has only recently been studied for large language models, where compression can significantly reduce coverage while leaving accuracy unchanged [10]; to our knowledge, no equivalent study exists for real-time object detectors on edge hardware.

This paper addresses this gap. Building directly on the calibration study in [1], we investigate whether YOLOv8n detectors exported at FP32, FP16, and INT8 via Intel OpenVINO maintain the nominal conformal coverage guarantee, both on clean test images and under the same controlled degradations studied previously.

Contributions. The main contributions of this work are:

- We present, to our knowledge, the first empirical study of split conformal prediction applied to a quantised edge-deployed object detector (YOLOv8n, FP32/FP16/INT8 via OpenVINO) on COCO data.
- We design a 24-condition evaluation matrix (3 precisions $\times$ 8 input conditions) on a held-out subset of COCO val2017 mirroring the degradation pipeline of [1], including Gaussian blur, JPEG compression, and additive Gaussian noise.
- We show that FP16 and INT8 quantised models fail to meet the nominal 90% coverage target on clean data and violate coverage in 6/8 and 3/8 conditions respectively, compared to 1/8 for FP32. Throughout this work, coverage is evaluated exclusively over matched detections ($\text{IoU} \geq 0.5$) and should be interpreted as a conditional reliability measure, not an end-to-end object detection guarantee.
- We provide an empirical reliability interpretation of confidence convergence, connecting Phase-I findings to conformal nonconformity score distributions and demonstrating that tail behaviour, not mean score alone, determines coverage.

II. RELATED WORK

A. Calibration and Confidence in Quantised Detectors

The calibration of deep neural networks was first systematically studied by Guo et al. [15], who showed that modern classifiers are often overconfident and proposed temperature scaling and Expected Calibration Error (ECE) as practical diagnostics. Subsequent work extended calibration analysis to object detection, where confidence scores jointly encode objectness and class probabilities [16]. In [1] we applied these tools to a quantised YOLOv8n detector and introduced the confidence convergence phenomenon, showing that quantisation-induced confidence gaps on clean data collapse under degradation.

B. Conformal Prediction

Conformal prediction provides distribution-free prediction sets with finite-sample marginal coverage guarantees under exchangeability [3]. Split conformal prediction [4] computes a nonconformity threshold on a held-out calibration set, then

uses it to accept or reject test predictions. Angelopoulos and Bates [2] provide a widely used introduction covering LAC scores, adaptive prediction sets (APS), and RAPS. Theoretical analysis shows that coverage can degrade under violations of exchangeability, including covariate and label shift [9].

C. Conformal Prediction for Object Detection

Timans et al. [5] develop a two-step conformal approach that propagates class-label uncertainty into bounding-box coverage intervals, validated on 2D localisation benchmarks. Copley et al. [6] study conformal methods for detection in aerial imagery. And  ol et al. [7] introduce sequential risk control for detection providing guarantees on false positive and false negative rates. These works operate at full precision on clean test images and do not consider quantised or degradation-exposed settings.

D. Conformal Prediction Under Distribution Shift and Compression

Tibshirani et al. [9] show that conformal coverage degrades under covariate shift, establishing the theoretical risk of deploying CP-calibrated models outside their training distribution. Angelopoulos et al. [11] address this directly by introducing adaptive conformal inference, which dynamically adjusts the nonconformity threshold to maintain coverage under distribution shift. Tong et al. [10] benchmark the effect of compression on conformal coverage in large language models, showing that quantisation and sparsity can cause sharp coverage declines without accuracy loss. Our work extends this analysis to real-time edge object detection for the first time.

III. METHODOLOGY

A. Model and Deployment Setup

We study a YOLOv8n detector [12] with pre-trained COCO weights, without fine-tuning. The model is exported to Intel OpenVINO [13] in three precision configurations: FP32 (full-precision baseline), FP16 (half-precision), and INT8 (post-training quantisation using NNCF). INT8 calibration uses the COCO128 training subset as the calibration data for activation scale estimation [17]. No quantisation-aware retraining is performed; all three models share identical pre-trained weights prior to export, ensuring that behavioural differences arise from precision changes alone. All experiments are conducted on a consumer-grade laptop with an Intel Core i5 CPU and 16 GB RAM, batch size 1, input resolution 640×640 pixels. On this hardware, the three precision configurations achieve inference throughputs of approximately 5.5 FPS (FP32), 10.1 FPS (FP16), and 19.0 FPS (INT8), consistent with the throughput characterisation reported in [1].

B. Datasets and Splits

We use two COCO-based datasets:

- **COCO128** [14]: A curated 128-image subset of the MS COCO dataset with dense annotations. We use 126 images with valid labels from the `train2017` split, constructing a 70/30 split and using the 70% portion

(approximately 88 images) exclusively for conformal calibration.

- **COCO val2017** [14]: The standard 5,000-image validation set. We download the official archive, convert annotations to YOLO format, and randomly sample 200 images with non-empty labels as the held-out test set (fixed random seed for reproducibility).

The key design choice is that INT8 quantisation uses COCO128 for calibration, while conformal test coverage is evaluated exclusively on COCO val2017. This avoids data leakage: the test distribution for conformal prediction does not overlap with the OpenVINO INT8 calibration data.

C. Nonconformity Score and Conformal Threshold

We adopt the Least Ambiguous Classifier (LAC) nonconformity score [2]. For each predicted detection matched to a ground-truth bounding box with Intersection-over-Union (IoU) ≥ 0.5 , we define:

$$s(x, y) = 1 - \hat{\pi}(y | x), \quad (1)$$

where $\hat{\pi}(y | x)$ is the detector’s confidence score for the matched class. False positives (predictions with IoU < 0.5) are excluded from calibration and coverage metrics, as they do not correspond to a well-defined ground truth and would otherwise trivialise the conformal threshold. Predictions were retained using a confidence threshold of 0.25 and non-maximum suppression (NMS) IoU threshold of 0.45 (Ultralytics defaults); a predicted box was then matched to a ground-truth annotation using greedy IoU matching at IoU ≥ 0.5 with class-label agreement required, and each ground-truth object was matched to at most one prediction; unmatched predictions and unmatched ground-truth objects were excluded from all conformal calculations. Matched detections are treated as the exchangeable unit for conformal prediction, as each represents a model prediction paired with a verified ground-truth annotation under a fixed IoU threshold. We treat matched detections as approximately exchangeable evaluation units under this fixed protocol, acknowledging that correlations within images may violate strict exchangeability in practice [3].

The LAC score is selected for three reasons. First, it is the canonical nonconformity score for classification tasks and is derived directly from the softmax confidence output already produced by YOLOv8n, requiring no architectural modification [2]. Second, LAC produces singleton or near-singleton prediction sets under high confidence, which is operationally appropriate for single-label object class prediction in detection pipelines. Third, its direct correspondence to detector confidence enables the connection to the confidence convergence phenomenon studied in [1]: a score defined as $1 - \hat{\pi}$ maps coverage violations directly back to confidence distribution shifts, making the cross-paper narrative internally consistent. Alternative scores such as APS and RAPS [2] are better suited to multi-label or ambiguous classification settings and introduce additional tuning parameters; their evaluation under quantisation is deferred to future work.

TABLE I
CLEAN TEST COVERAGE ON COCO val2017. TARGET: CR ≥ 0.90 .

Precision	Det.	$\hat{q}$	Coverage Rate	95% Wilson CI
FP32	695	0.6771	0.9165 ✓	[0.8936, 0.9348]
FP16	700	0.6628	0.8929 ×	[0.8678, 0.9137]
INT8	703	0.6670	0.8976 ×	[0.8730, 0.9179]

Given calibration scores $\{s_i\}_{i=1}^n$, we compute the conformal threshold $\hat{q}$ at error level $\alpha = 0.10$ using the finite-sample-corrected quantile level:

$$\gamma = \frac{\lceil (n+1)(1-\alpha) \rceil}{n}, \quad (2)$$

setting $\hat{q} = \text{Quantile}_\gamma(\{s_i\})$ with the “higher” interpolation method [3], [4]. At test time, a matched detection is accepted into the conformal prediction set if $s \leq \hat{q}$; under exchangeability, the marginal coverage rate should be at least $1 - \alpha = 0.90$ [2]. Separate thresholds $\hat{q}_{\text{FP32}}$, $\hat{q}_{\text{FP16}}$, and $\hat{q}_{\text{INT8}}$ are computed using the same calibration images but precision-specific model predictions.

D. Input Degradation Pipeline

Following the degradation protocol of [1], each of the 200 COCO val2017 test images is processed under seven degraded conditions: Gaussian blur with $\sigma \in \{1, 2, 3\}$; JPEG compression with quality factors $Q \in \{90, 70, 50\}$; and additive Gaussian noise with zero mean and standard deviation $\sigma = 25$. This yields eight conditions in total per precision level (clean + seven degraded). Degraded images share the same annotation files as clean images since degradation affects only pixel values, not object locations.

E. Metrics and Statistical Analysis

For each (precision, condition) pair we compute the empirical **coverage rate** (CR): the fraction of matched test detections for which $s \leq \hat{q}$. A **coverage violation** is recorded when $\text{CR} < 0.90$. We also track mean and standard deviation of nonconformity scores to study distributional convergence across precisions.

Pairwise Mann–Whitney U tests are performed on calibration nonconformity scores between all three precision pairs (FP32/FP16, FP32/INT8, FP16/INT8), with rank-biserial correlation as effect size, to test whether quantisation induces statistically significant shifts in the calibration residuals.

IV. EXPERIMENTS

A. Calibration on COCO128

Table II summarises conformal calibration results on the COCO128 calibration split. All three precision levels yield similar statistics: the FP32, FP16 and INT8 models produce 345, 343 and 347 matched detections, with mean confidence values of 0.6323, 0.6351 and 0.6399 respectively. Mean nonconformity scores are 0.3677, 0.3649 and 0.3601 with standard deviations around 0.21. The resulting conformal thresholds

TABLE II
CONFORMAL CALIBRATION SUMMARY ON COCO128 ($\alpha = 0.10$).

Precision	Det.	Mean Conf.	Mean s	Std s	$\hat{q}$
FP32	345	0.6323	0.3677	0.2120	0.6771
FP16	343	0.6351	0.3649	0.2086	0.6628
INT8	347	0.6399	0.3601	0.2128	0.6670

at $\alpha = 0.10$ are $\hat{q}_{\text{FP32}} = 0.6771$, $\hat{q}_{\text{FP16}} = 0.6628$, and $\hat{q}_{\text{INT8}} = 0.6670$.

Quantisation thus shifts thresholds only modestly; FP16 exhibits the tightest threshold, consistent with its slightly higher average calibration confidence. Pairwise Mann–Whitney U tests on calibration nonconformity scores reveal no statistically significant differences across any precision pair (FP32 vs. FP16: $U = 59,614$, $p = 0.864$, $r = -0.008$; FP32 vs. INT8: $U = 61,231$, $p = 0.602$, $r = -0.023$; FP16 vs. INT8: $U = 60,465$, $p = 0.716$, $r = -0.016$). The negligible effect sizes confirm that quantisation shifts are too subtle to appear in aggregate calibration statistics, yet are sufficient to alter upper-tail behaviour and consequently break coverage — reinforcing that tail mass above $\hat{q}$, not distributional mean shift, is the mechanism driving violations.

B. Clean Coverage on COCO val2017

Table I reports coverage on the 200-image COCO val2017 test set without degradation. The FP32 model attains an empirical coverage rate of **0.9165** across 695 matched detections, meeting the 90% target. In contrast, FP16 and INT8 achieve 0.8929 (700 detections) and 0.8976 (703 detections) respectively, both falling below the nominal 90% coverage target on the clean test subset. The corresponding miscoverage rates are approximately 10.7% and 10.2% for FP16 and INT8 — consistent with quantisation-aware CP miscoverage reported in other domains [10]. These results are consistent with the finite-sample coverage guarantee under exchangeability [3]: the LAC threshold $\hat{q}$ is calibrated to satisfy $\text{CR} \geq 1 - \alpha$ with high probability when the exchangeability assumption holds, and the observed FP32 coverage of 0.9165 confirms this guarantee is met at full precision.

C. Coverage Under Input Degradation

Table III presents the full 24-condition coverage matrix. FP32 maintains coverage ≥ 0.90 in seven of eight conditions, with a single violation under additive noise ($\text{CR} = 0.8956$). FP16 violates coverage in six conditions (clean, blur $\sigma = 1$, blur $\sigma = 3$, JPEG $Q = 90$, JPEG $Q = 70$, noise). INT8 violates coverage in three conditions (clean, JPEG $Q=90$, and noise). Counting violations per precision: FP32 1/8, FP16 6/8, INT8 3/8.

Representative examples: under blur $\sigma = 3$, coverage rates are 0.9178 (FP32), 0.8835 (FP16), and 0.9165 (INT8); under JPEG $Q = 70$ they are 0.9139, 0.8877, and 0.9103 respectively. FP16 is thus the least reliable precision from a formal coverage perspective, even though it is commonly assumed to be a “safe” intermediate option.

TABLE III
EMPIRICAL COVERAGE RATE (CR) ACROSS ALL 24 CONDITIONS. BOLD VALUES INDICATE VIOLATIONS ($\text{CR} < 0.90$).

Condition	Type	FP32	FP16	INT8
Clean	—	0.9165	0.8929	0.8976
Blur $\sigma = 1$	Blur	0.9051	0.8970	0.9131
Blur $\sigma = 2$	Blur	0.9275	0.9138	0.9138
Blur $\sigma = 3$	Blur	0.9178	0.8835	0.9165
JPEG $Q = 90$	JPEG	0.9006	0.8910	0.8881
JPEG $Q = 70$	JPEG	0.9139	0.8877	0.9103
JPEG $Q = 50$	JPEG	0.9230	0.9022	0.9130
Noise	Noise	0.8956	0.8977	0.8965
Violations		1/8	6/8	3/8

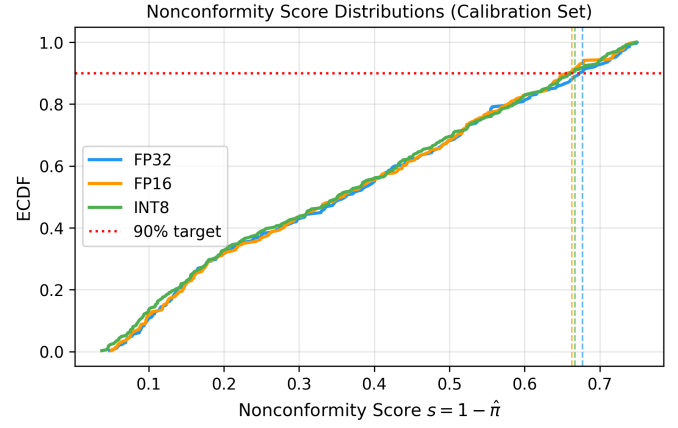


Fig. 1. Empirical cumulative distribution of LAC nonconformity scores on the COCO128 calibration set for FP32, FP16, and INT8. Dashed vertical lines mark each model’s conformal threshold $\hat{q}$. The horizontal dotted line marks the 90% coverage target. Tail differences above $s = 0.65$ drive the empirical coverage violations reported in Table III.

D. Nonconformity Score Convergence

On the clean test set, mean nonconformity scores are 0.3414 (FP32), 0.3410 (FP16), and 0.3353 (INT8). Under degradation these means converge: for example, under blur $\sigma = 2$ the spread is 0.3443–0.3467 across precisions, and under JPEG $Q = 50$ it is 0.3367–0.3495. This mirrors the confidence convergence phenomenon in [1]. However, despite convergence in means, FP16 exhibits more frequent violations, indicating that upper-tail differences in nonconformity distributions — not averages — are the determinants of coverage. To support this, we note that the 90th-percentile nonconformity scores on the calibration set are 0.677 (FP32), 0.663 (FP16), and 0.667 (INT8), closely tracking the estimated thresholds $\hat{q}$ and confirming that coverage behaviour is driven by tail mass above $\hat{q}$ rather than by shifts in the score mean.

V. DISCUSSION

RQ1 — Threshold shifts. Conformal thresholds differ only modestly across precisions ($\hat{q}$ range: 0.6628–0.6771), confirming that quantisation acts as a fine perturbation of calibration residuals rather than a dramatic rescaling. FP16 has the tightest threshold, yet exhibits the worst coverage.

RQ2 — Clean coverage. Despite similar thresholds, FP16 and INT8 fail the nominal 90% target on clean data. This shows that even small systematic shifts in the nonconformity score distribution, too subtle to appear in accuracy or ECE metrics, are sufficient to undermine the intended nominal coverage target [11], [10]. However, the 95% Wilson confidence intervals for FP16 [0.8678, 0.9137] and INT8 [0.8730, 0.9179] both straddle the 0.90 threshold, indicating that the observed violations are not statistically conclusive at the 95% confidence level and should be interpreted as indicative rather than definitive findings. Larger calibration and test sets would be required to confirm these violations with high statistical certainty.

RQ3 — Degradation robustness. FP32 is the most robust, FP16 the worst, and INT8 intermediate. The combination of quantisation and input degradation disproportionately harms FP16 coverage, consistent with CP sensitivity to distribution shift reported in [9], [11]. For practitioners, precision choice cannot be made on throughput grounds alone when empirical reliability is required.

RQ4 — Formalising confidence convergence. Mean nonconformity scores converge under degradation, echoing [1]. Together, the two studies present a coherent narrative: quantisation changes how confident the model sounds (Phase 1), and these changes, combined with distribution shift, are sufficient to undermine empirical marginal coverage guarantees under exchangeability (Phase 2). Conformal prediction thus provides a principled audit tool for compressed models. Fig. 1 confirms that upper-tail differences in nonconformity distributions — not mean scores — are the primary driver of coverage violations, as FP16 accumulates more mass above $\hat{q}$ than FP32 despite near-identical means.

VI. LIMITATIONS AND FUTURE WORK

The calibration set (COCO128, ~ 88 images) and the 200-image test subset limit the granularity of coverage estimates. The study covers a single architecture (YOLOv8n) and one dataset family (COCO). We use standard split CP with LAC scores; two-step or object-wise conformal schemes [5], [6] may behave differently under quantisation.

To provide a well-defined basis for these extensions, we outline three directions whose formulations are established here and whose full implementations are deferred to future work.

Joint nonconformity score. The LAC score $s(x, y) = 1 - \hat{\pi}(y|x)$ captures classification confidence but ignores spatial misalignment. A natural extension is a joint score of the form

$$s(x, y) = \alpha(1 - \hat{\pi}(y|x)) + (1 - \alpha)(1 - \text{IoU}),$$

which penalises both poor classification and bounding-box imprecision simultaneously. Calibrating such a score under quantisation, and selecting α in a data-driven manner, is left for future work.

Conformal Risk Control (CRC). The current framework excludes false positives from calibration and coverage, restricting analysis to matched detections. Conformal Risk Control [8] provides a principled alternative: by defining a loss function

over arbitrary detection errors (e.g., false discovery rate or false positives per image), CRC extends formal coverage guarantees to the full detection output including unmatched predictions. Applying CRC to quantised edge detectors under degradation is a direct avenue for future work.

Adaptive Conformal Inference (ACI). Under input degradation, the exchangeability assumption underlying split CP is violated, causing empirical coverage to fall below the nominal target. Adaptive Conformal Inference [11] addresses this by maintaining a rolling estimate of recent coverage and dynamically adjusting the error level α_t to compensate for distribution shift. Integrating ACI into the edge deployment pipeline to provide online coverage maintenance under sensor degradation is a natural next step.

Future work will also extend this analysis to larger datasets, additional architectures (e.g., RT-DETR), quantisation-aware training, and additional CP variants, integrating calibration and coverage into a unified reliability framework for compressed edge detectors.

VII. CONCLUSION

We have presented the first empirical study of split conformal prediction applied to a quantised edge-deployed YOLOv8n object detector. A 24-condition evaluation on COCO val2017 showed that FP32 largely maintains the intended 90% coverage target, while the empirical coverage point estimates for FP16 and INT8 fall below the nominal 90% target on clean test subset, with coverage violations in 6/8 and 3/8 degradation conditions respectively. Coverage failures are connected to the confidence convergence phenomenon reported in [1], demonstrating that tail differences in nonconformity distributions — not means — determine empirical reliability. Post-training quantisation must be treated as a model change: conformal guarantees require re-validation after compression, and coverage under realistic degradation conditions should be part of the model selection process for latency-sensitive edge deployments where empirical reliability under compression must be validated.

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